Hybrid Reinforcement Learning and Genetic Algorithms for Meta-Learning Audio Denoising

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Paper version v4 (introduction + literature landscape; experiments ongoing)

Code: https://github.com/reeldemo/reelsynth, https://github.com/reeldemo/denoise-opt-meta

Abstract

Periodic discontinuities in cyclic audio frames inject broadband crackle under wrap-around playback. We treat cycle-local seam restoration as an instance of audio signal denoising and search denoising operators with a hybrid meta-learning outer loop that interleaves genetic-algorithm population updates with reinforcement-learning architecture proposals, optionally with population-based exploit–mutate schedules and mixture-of-experts soft gates. Candidates are scored by a prolonged residual $R \in [0, 1]$ (1 = best) against an ideal multi-period tiling, without paired clean waveforms for every trial. This draft delivers the problem framing and a used-versus-screened literature landscape grounded on OA PDF analyses (Noise2Noise, PBT, Aging Evolution, ERL, ENAS/PPO, Wave-U-Net/Conv-TasNet, MoE, DDSP; Demucs/DiffWave/MAML/DARTS screened). A large GPU campaign (PPO+GA+PBT+NAS) is ongoing; final quantitative tables are deferred until the declared budget completes.

Keywords audio denoising · neural architecture search · reinforcement learning · genetic algorithms · wavetable synthesis · meta-learning

1 Introduction

Audio denoising asks how to recover a usable signal from observations that are corrupted by structured or unstructured interference. In many practical settings the corruption is not additive white noise but a deterministic, periodically reinforced artifact: a discontinuity that reappears every period under cyclic playback. Wavetable and related one-cycle synthesizers make this concrete. A single-period frame is stored and phase wraps modulo one; if the sample values at the two ends disagree, the wrap injects a broadband click that repeats at the fundamental—audible as seam crackle. The same geometry appears whenever a short cyclic buffer is looped, cross-faded poorly, or tiled for evaluation. We therefore treat *cycle-local seam restoration* as an instance of audio signal denoising: the operator edits a short neighborhood of the wrap (and optionally a lightweight residual cell over the cycle) so that prolonged tiled playback agrees with an ideal reference that withholds the open-wrap cliff.

Classical virtual-analog and anti-aliasing practice already supplies useful local mechanisms—cosine crossfades, BLEP/BLAMP-style discontinuity control, and related oscillator design [1, 2, 3]—yet edited frames still need an *outer* score that measures prolonged agreement, not only a seam-local energy proxy. Label-free restoration philosophies such as Noise2Noise show that useful reconstructors can be trained without paired clean targets when the evaluation geometry is carefully defined [4]. On the architecture side, surveys of neural architecture search (NAS) [10], reinforcement-learning controllers with parameter sharing [11], aging/regularized evolution [13], population-based hyperparameter schedules [14], and evolution-guided policy gradients [15] collectively argue that discrete search over operators benefits from *hybrid* outer loops: population diversity from genetic algorithms (GA), directed proposals from reinforcement learning (RL), and exploit–mutate schedules from population-based training (PBT). Proximal policy optimization (PPO) offers a stable clipped surrogate for proposing discrete edits [12]; sparsely gated mixture-of-experts (MoE) layers motivate soft routing among heterogeneous blocks [16]. Waveform and time-domain separation networks such as Wave-U-Net and Conv-TasNet [7, 9], together with broader deep audio surveys [6], inform searchable cell families (tiny U-Net / dilated / attention blocks) without requiring full production separators at every trial.

The gap we address is not "yet another end-to-end denoiser," but a *meta-learning* protocol for this periodic instance: an outer hybrid RL+GA search that proposes denoising architectures and bake parameters, scored by a prolonged residual $R \in [0, 1]$ (1 = best) against an ideal multi-period tiling, without assuming paired studio-clean waveforms for every candidate. Differentiable DSP modules [5] situate seam operators in an interpretable audio stack; Bayesian and multi-objective evolutionary HPO [17, 18] inform prior families that compete under the same residual objective. We deliberately avoid claiming a finished large-scale leaderboard in this draft: a GPU overnight run (PPO+GA+PBT+NAS with depth and MoE biases) is still in progress. Early checkpoints already exceed historical DualCosine baselines on residual, but final budgets and held-out matrices will be reported only after completion.

Contributions.

1. **Problem framing.** Cast periodic wrap crackle / cycle-local seam restoration as audio signal denoising under prolonged tiled evaluation, separating seam-local diagnostics from a residual score R.

2. **Hybrid meta-outer loop.** Specify an RL+GA(+PBT) search over discrete operator graphs and continuous bake parameters, with literature-grounded branch names (GA crossover, PPO mutation, PBT exploit, MoE soft gates) rather than undifferentiated "AutoML."

3. **Literature protocol.** Organize used versus screened prior work for audio denoise architectures and RL–evolutionary hybrids, grounded on attached OA PDFs where available.

4. **Open evaluation path.** Release companion code for the synthesizer engine and meta search; report interim protocol honestly while large-scale experiments finish.

Section 2 surveys the literature landscape. Sections 3–4 outline the residual objective and search protocol (placeholders pending final ingest). Results and discussion will be completed when the overnight campaign reaches its declared budget.

2 Literature Landscape

We organize prior work by *mechanism*, not chronology. The goal is to show which lines of research our residual-scored RL+GA meta-search depends on (**used**), and which high-visibility lines we inspected but do not treat as methodological dependencies (**screened**). Paraphrases below prefer attached abstracts and OA PDF snippets over title-only name-dropping.

2.1 Classical discontinuity control and modular audio DSP

Virtual-analog oscillator and filter design supplies the classical vocabulary for wrap and discontinuity artifacts [1]. Aliasing reduction for clipped and discontinuous waveforms, and BLAMP-style corner rounding, motivate seam-local corrections that respect band-limited intent [2, 3]. Differentiable DSP (DDSP) reframes such modules as composable, interpretable blocks inside learning pipelines [5]; we use that framing for *small* bake/FIR/MLP seam cells rather than end-to-end neural vocoders. These works are **used** as domain anchors for the operator vocabulary and for why prolonged cyclic playback is the right outer judge.

2.2 Label-free and unsupervised restoration

Noise2Noise shows that restoration mappings can be learned from corrupted observations alone, without clean targets, by exploiting statistical structure of the corruption process [4]. That philosophy motivates scoring denoise candidates without paired "studio-clean" wavetable cycles: our residual compares an engine-baked tiling to a procedural ideal that withholds open-wrap cliffs, not to a human-cleaned master. We **use** Noise2Noise as conceptual justification for unsupervised residual ranking; we do *not* claim to retrain a Noise2Noise CNN on images.

2.3 Waveform denoising and separation architectures

Deep learning for audio surveys the move from hand-designed features to learned time/frequency models [6]. Wave-U-Net performs end-to-end source separation with multi-scale 1-D U-Nets [7]; singing-voice U-Nets similarly popularized encoder–decoder skips on audio [8]. Conv-TasNet demonstrated strong time-domain separation with temporal convolutional architectures [9]. These lines are **used** as design priors for *tiny* searchable cells (shallow U-Net / dilated / dual-path / attention blocks under a strict per-trial bake budget). They are not used as frozen full-scale separators inside every meta trial.

2.4 Neural architecture search and RL controllers

Elsken et al. taxonomize NAS by search space, strategy, and evaluation cost [10]. ENAS couples an RL controller with parameter sharing for efficient discrete search [11]. We **use** this family as the conceptual ancestor of RL proposing architecture edits (add/remove/mutate ops, toggle residual-primary, mutate MLP/FIR), while scoring with prolonged residual rather than ImageNet reward. PPO supplies the clipped-surrogate policy update used when the overnight loop logs PPO mutation branches [12].

2.5 Evolutionary NAS and population schedules

Regularized / aging evolution showed that evolutionary NAS can discover competitive image classifiers when population turnover is carefully regularized [13]. Population Based Training jointly adapts weights and hyperparameters in an asynchronous population, discovering schedules rather than a single fixed hyperparameter vector [14]. Practical Bayesian optimization [17] and decomposition-based multi-objective evolution (MOEA/D) [18] further inform local Bayesian and multi-objective prior families in earlier DenoiseOpt searches. These are **used** for GA tournament/crossover branches, PBT-style exploit–mutate, and literature-combo priors—implemented at synthesizer-trial scale, not as distributed datacenter PBT.

2.6 Evolutionary reinforcement learning hybrids

Khadka and Tumer's evolution-guided policy gradient (ERL) interleaves an evolutionary population with an RL actor to mitigate sparse reward, weak exploration, and brittle hyperparameters in deep RL [15]. We **use** ERL as the *spirit* of interleaving GA population steps with PPO updates at tiny population sizes; we do not claim a full ERL robotics stack.

2.7 Mixture-of-experts soft gating

Sparsely gated MoE layers increase capacity via conditional computation [16]. We **use** MoE as motivation for soft gates over heterogeneous seam blocks (e.g., U-Net vs dilated vs attention cells) under residual scoring—again at bake-scale widths, not trillion-parameter routing.

2.8 Used versus screened

Used (methodological or comparative anchors).

- Discontinuity / VA lineage: [1, 2, 3].
- Unsupervised restoration philosophy: [4].
- Modular / differentiable audio DSP context: [5].
- Audio DL and compact waveform cells: [6, 7, 8, 9].
- NAS / RL controllers / PPO: [10, 11, 12].
- Evolutionary NAS and population HPO: [13, 14, 17, 18].
- ERL hybrid outer loop: [15].
- MoE soft gating prior: [16].

Screened (relevant contrast; not dependencies).

- **MAML** [19]: inspirational bi-level / fast-adaptation framing; we do *not* differentiate through a deep task network. Inner fits are coordinate-style updates on bake parameters; outer rank is residual R.
- **DARTS** [20]: continuous relaxation of architecture search; we keep discrete ops + small cells under residual scoring.
- **Demucs / Hybrid Demucs** [21, 22]: strong music separators, but full multi-scale / hybrid STFT stacks are too heavy per meta trial; screened for overnight bake budgets.
- **DiffWave** [23]: powerful diffusion vocoder; screened as a generative sampling stack, not a seam bake operator.
- **SEGAN-style adversarial enhancement** [24]: optional tiny adversarial cues only; full GAN training loops screened out for per-trial cost.

Positioning. Relative to this landscape, the present work contributes (i) a prolonged residual objective for periodic seam denoising, (ii) an explicitly hybrid RL+GA(+PBT) meta-outer loop with honest branch naming, and (iii) a used/screened literature protocol tied to OA PDF analyses. We do not claim SOTA on general speech enhancement or music separation corpora; the primary claim is protocol-level: residual-scored meta-search for cycle-local audio denoise operators under a declared compute budget.

3 Methods

Placeholder — full methods will track the frozen overnight protocol after the declared iteration budget completes.

Residual score. Let y be the engine-baked cycle and $r^{\star}$ an ideal multi-period reference from the same seed with open-wrap cliffs withheld. With tiling length N (typically N=16),

$$R = \mathrm{clamp}\left(1 - \frac{\mathrm{rms}(y_{\mathrm{tiled}} - r^{\star}_{\mathrm{tiled}})}{\max(\mathrm{rms}(r^{\star}_{\mathrm{tiled}}), \varepsilon)}, 0, 1\right). \tag{1}$$

By construction $R \in [0, 1]$ with 1 best. Soft shape gates (when enabled) down-weight mid-cycle damage.

Searchable operator. Candidates combine classical seam ops (e.g., dual-cosine, polish, soft seam, FIR blends) with tiny residual cells (MLP/FIR/U-Net-lite/attention-lite) and optional MoE soft gates. Bake parameters θ live in a bounded box; architecture strings are discrete.

Hybrid outer loop. Each iteration may (i) run GA tournament selection with crossover/mutation, (ii) sample PPO actions that edit ops/hyperparameters, (iii) apply PBT-style exploit–mutate on elites, and (iv) evaluate lit-combo priors. Selection is always by residual R (and logged auxiliaries). Implementation details and exact action sets will be frozen to the overnight run configuration in the results revision.

4 Experiments

Placeholder — protocol sketch only; numbers below are interim and must not be read as final.

Ongoing overnight campaign. A CUDA search tagged `PPO+GA+PBT+NAS+depth+MoE` is running with a large iteration budget (order 10^5). Early fitted champions already reach residual ≈ 0.987 on the training/validation geometry used by the runner, against historical DualCosine baselines near ≈ 0.70 on prior held-out matrices (v3 reported DualCosine residual ≈ 0.698 at N=2000). We will replace this paragraph with the completed budget, seeds, population size, and held-out matrix when the job finishes—without interrupting the live GPU process.

Prior completed study (v3 reference). A 1500-trial literature-informed CPU/GPU study selected `evo_explore_515` at residual ≈ 0.824 versus DualCosine ≈ 0.698 ($\Delta \approx +0.126$). That result anchors the residual-primary protocol; the overnight hybrid RL+GA run is the scale-up under test.

5 Results

Deferred for final claims. Final tables will be ingested after the overnight campaign reaches its declared iteration budget. Interim monitoring plots from an early GPU hybrid run are shown below for protocol transparency only; champion residuals and branch win-rates from incomplete runs are *not* asserted as finished science.

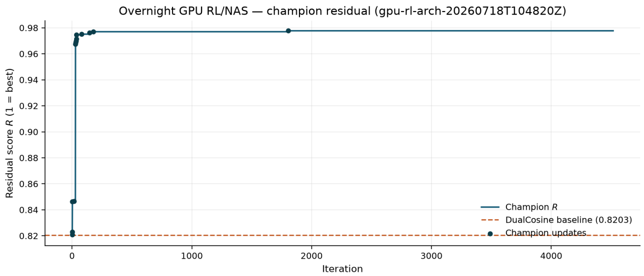

Figure 1: Interim champion residual R versus search iteration (monitoring snapshot). Not a final result.

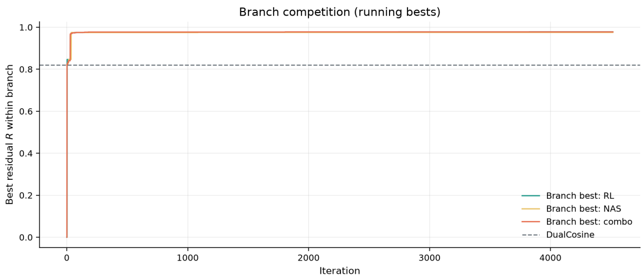

Figure 2: Interim per-branch best residual trajectories (GA / RL / PBT / NAS / combo). Monitoring only.

6 Discussion

Deferred until Results are complete. Anticipated themes: which hybrid branch (GA / PPO / PBT / lit-combo / MoE) actually wins under residual; whether deeper cells help when R holds; reconciliation with the v3 finding that light evolutionary explore beat nested loss-opt elites under a smaller budget.

7 Limitations

- Residual uses a procedural ideal (no-cliff sibling), not a perceptual MOS or listening panel.
- Bake metrics are not substitute for formal audio quality tests.
- Per-trial architecture width is capped for overnight throughput; full Demucs-/diffusion-scale stacks were screened out.
- Large-scale overnight results in this draft are interim; final claims require completed budgets and frozen seeds.
- Classical VA/BLEP citations are domain anchors from prior versions; not all have OA PDFs in the current harvest folder.

8 Conclusion

We framed periodic wrap crackle as an audio denoising instance and outlined a hybrid RL+GA meta-learning outer loop scored by prolonged residual R. The literature landscape separates used anchors (Noise2Noise, PBT, Aging Evolution, ERL, ENAS/PPO, Wave-U-Net/Conv-TasNet, MoE, VA/BLEP, DDSP) from screened contrasts (MAML, DARTS, Demucs, DiffWave, full GAN loops). Companion code lives at `https://github.com/reeldemo/reelsynth` and `https://github.com/reeldemo/denoise-opt-meta`. Final quantitative claims await completion of the ongoing GPU campaign.

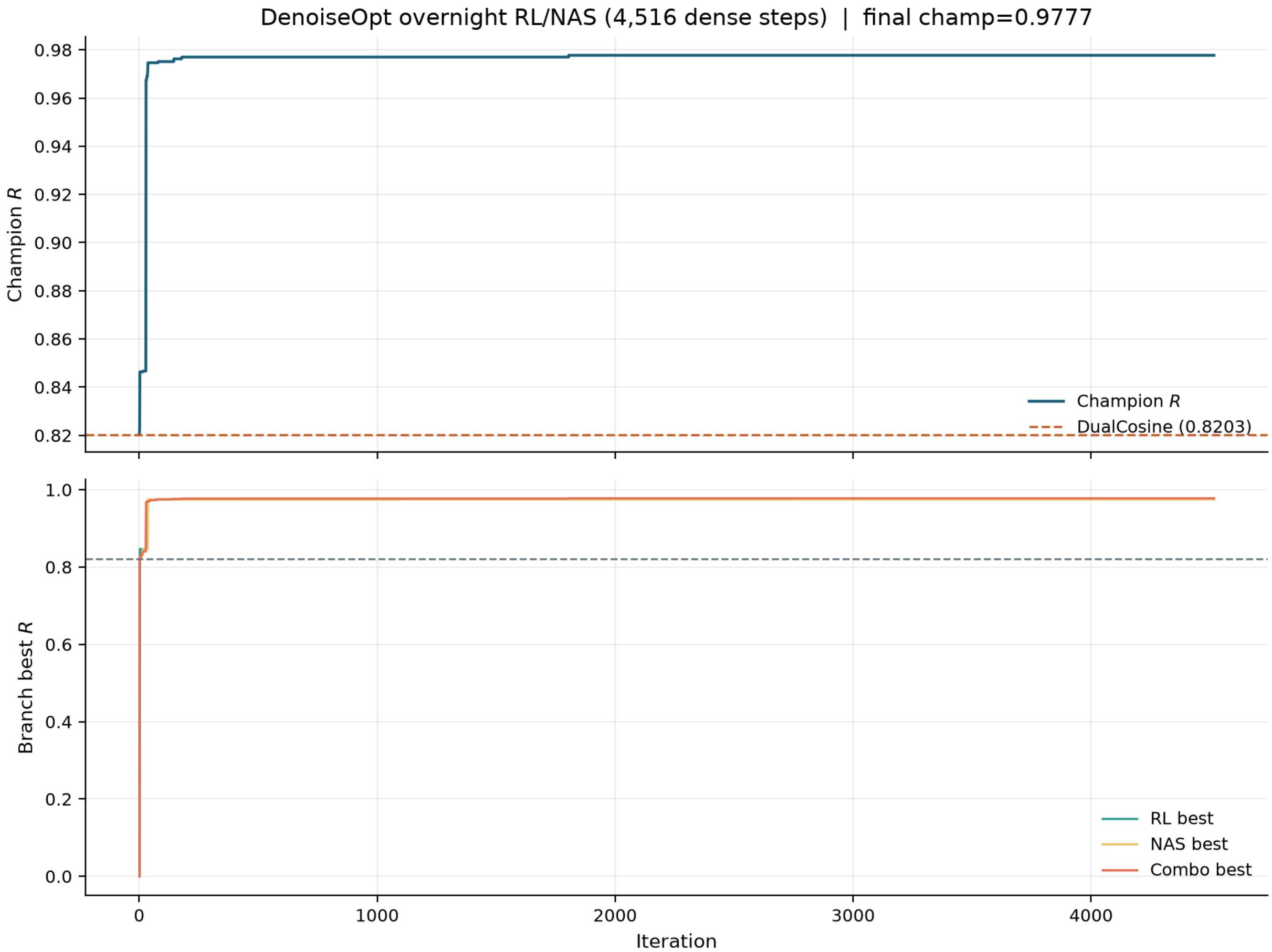

Figure 3: Full-width interim overnight monitoring panel (residual, branches, champion timeline). Snapshot for transparency while the GPU budget completes.

Figure 4: Interim residual distribution by search branch. Monitoring snapshot.

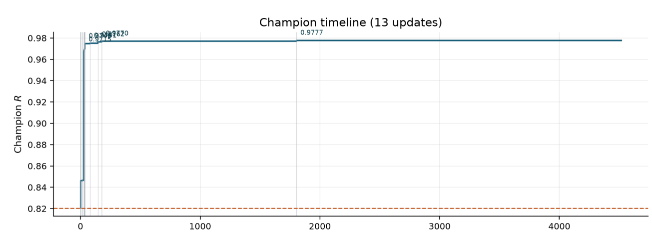

Figure 5: Interim champion update timeline (iteration, residual, branch). Monitoring only.

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